

Cycle Structure Generation

Completing Cycle Structures on Baca's Proposed Chains Based on Obligate and Most Probable Structure

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#1 Recap

What Baca et al. Found

Punctuated Evolution of Prostate Cancer Genomes

[Sylvan C. Baca](#)^{1,2,3} · [Davide Prandi](#)⁶ · [Michael S. Lawrence](#)² · ... · [Francesca Demichelis](#)^{6,9,18}   .
[Mark A. Rubin](#)^{8,11,18} · [Levi A. Garraway](#)^{1,2,3,4,18}   ... [Show more](#)

Summary

The analysis of exonic DNA from prostate cancers has identified recurrently mutated genes, but the spectrum of genome-wide alterations has not been profiled extensively in this disease. We sequenced the genomes of 57 prostate tumors and matched normal tissues to characterize somatic alterations and to study how they accumulate during oncogenesis and progression. By modeling the genesis of genomic rearrangements, we identified abundant DNA translocations and deletions that arise in a highly interdependent manner. This phenomenon, which we term “chromoplexy,” frequently accounts for the dysregulation of prostate cancer genes and appears to disrupt multiple cancer genes coordinately. Our modeling suggests that chromoplexy may induce considerable genomic derangement over relatively few events in prostate cancer and other neoplasms, supporting a model of punctuated cancer evolution. By characterizing the clonal hierarchy of genomic lesions in prostate tumors, we charted a path of oncogenic events along which chromoplexy may drive prostate carcinogenesis.

Baca et al. sequenced the complete genomes of 57 prostate tumors and found a

CHRONOLOGY of Oncogenic Events in Prostate Cancer Progression.

Clonality Concept

Mutations that happen **EARLY** are present in **EVERY** tumor cell and are called **clonal mutations**.

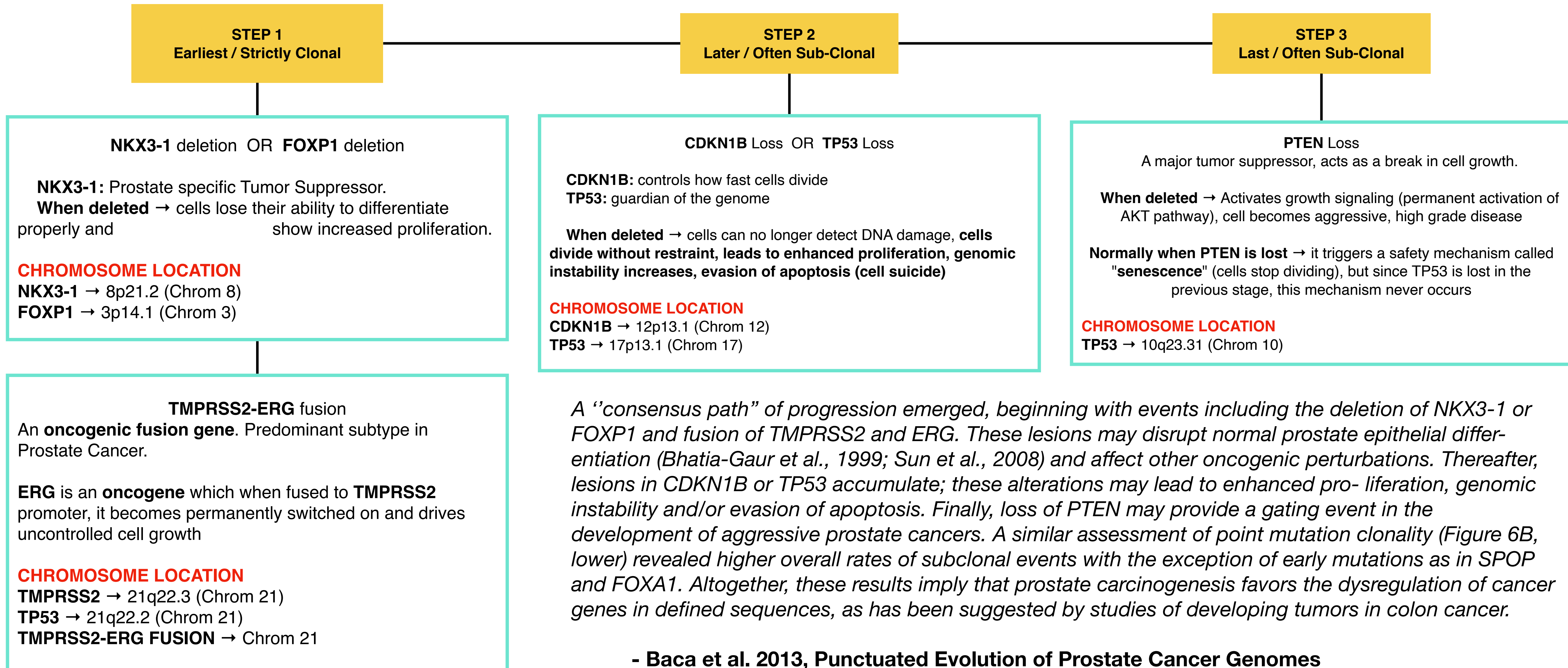
Mutations that happen **LATER** are present in only **SOME** tumor cells and are called **sub-clonal mutations**.

By measuring what fraction of tumor cells carry each mutation, Baca could reconstruct the timeline,

Clonal = happened before the tumor expanded = **EARLY**

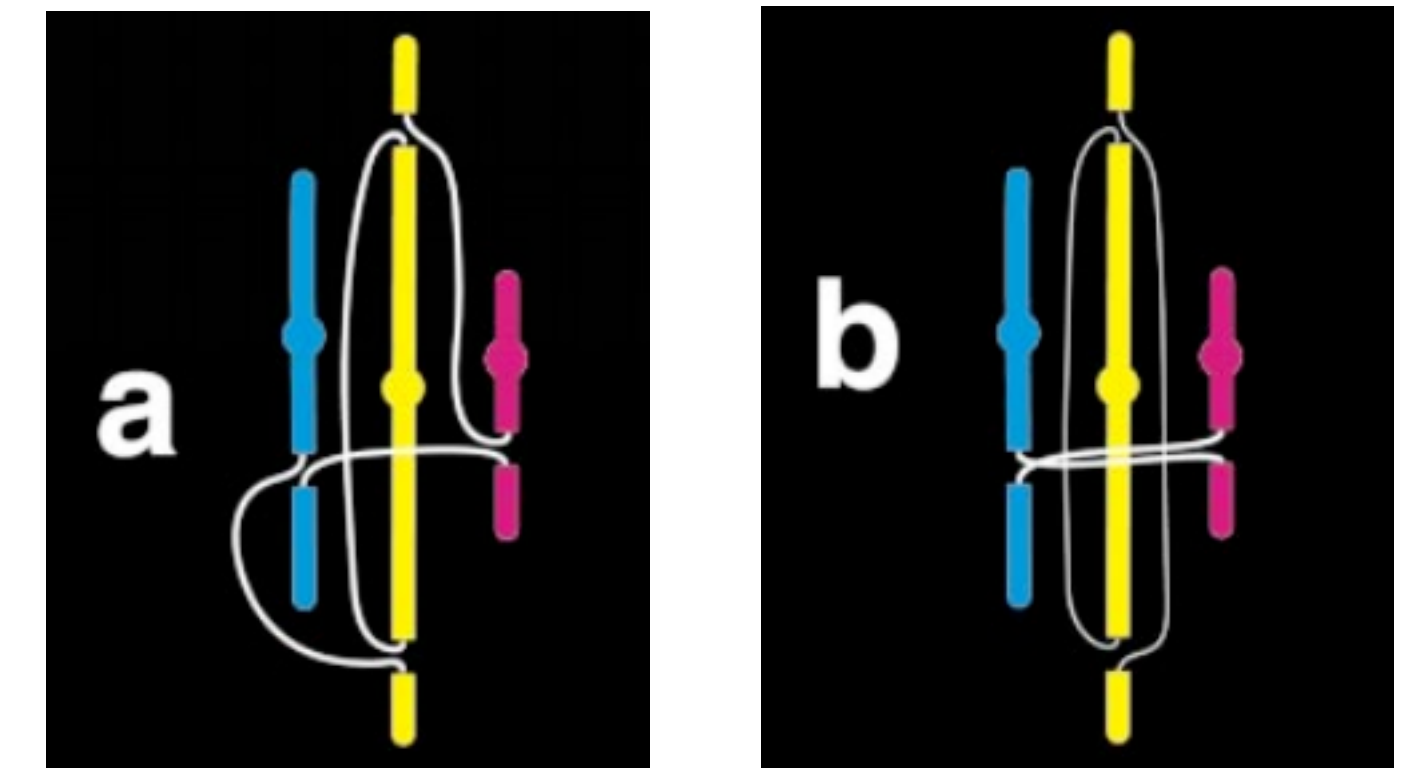
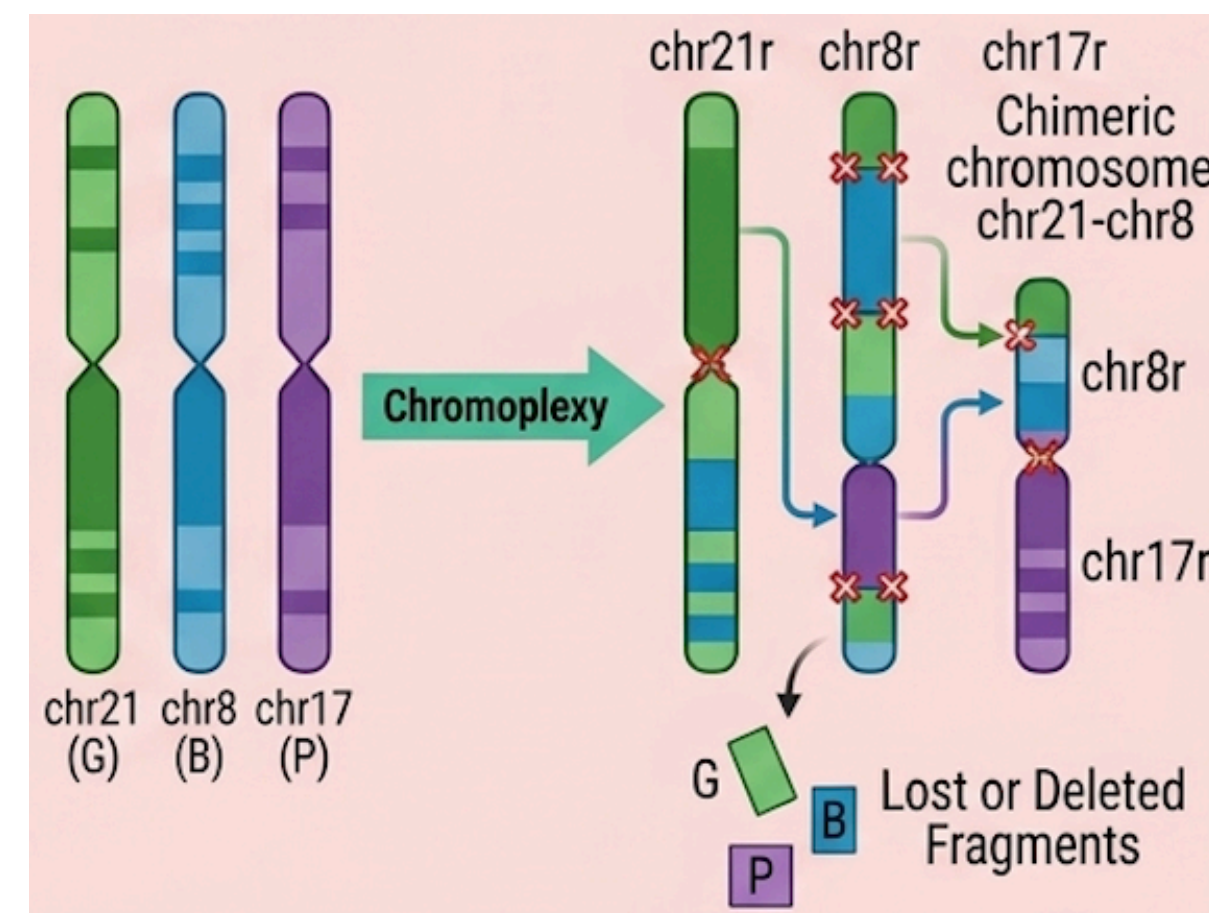
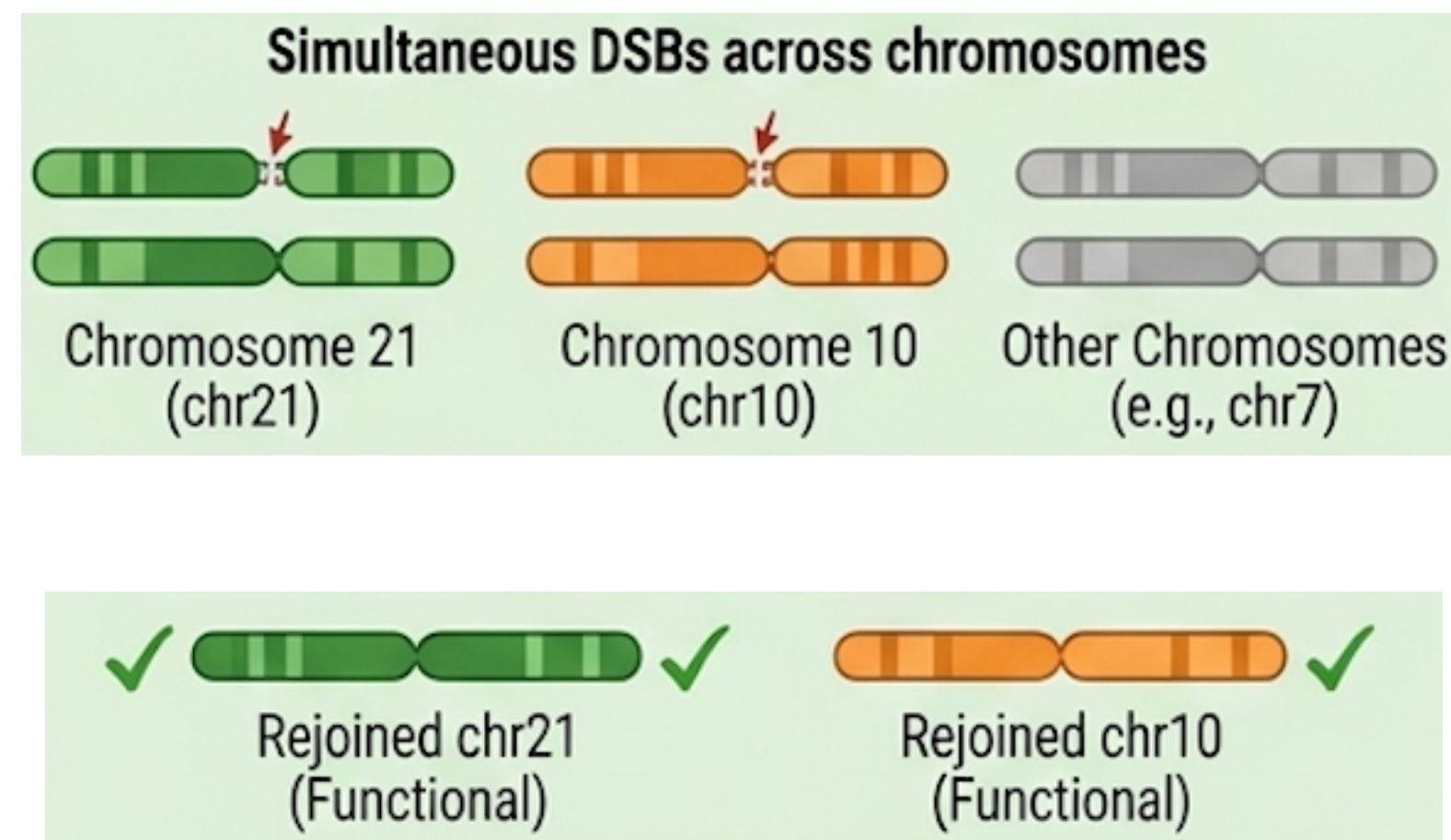
Subclonal = happened after the tumor was established = **LATE**

The Consensus Path



Chromoplexy: Coordinated Genome Restructuring

When multiple DNA **double strand breaks** occur **simultaneously** (**coordinated event**) **across different chromosomes**, in the **same small region** of the nucleus, instead of one at a time (**independent event**), the broken ends can rejoin each other in a chain which may differ from the original pairing.

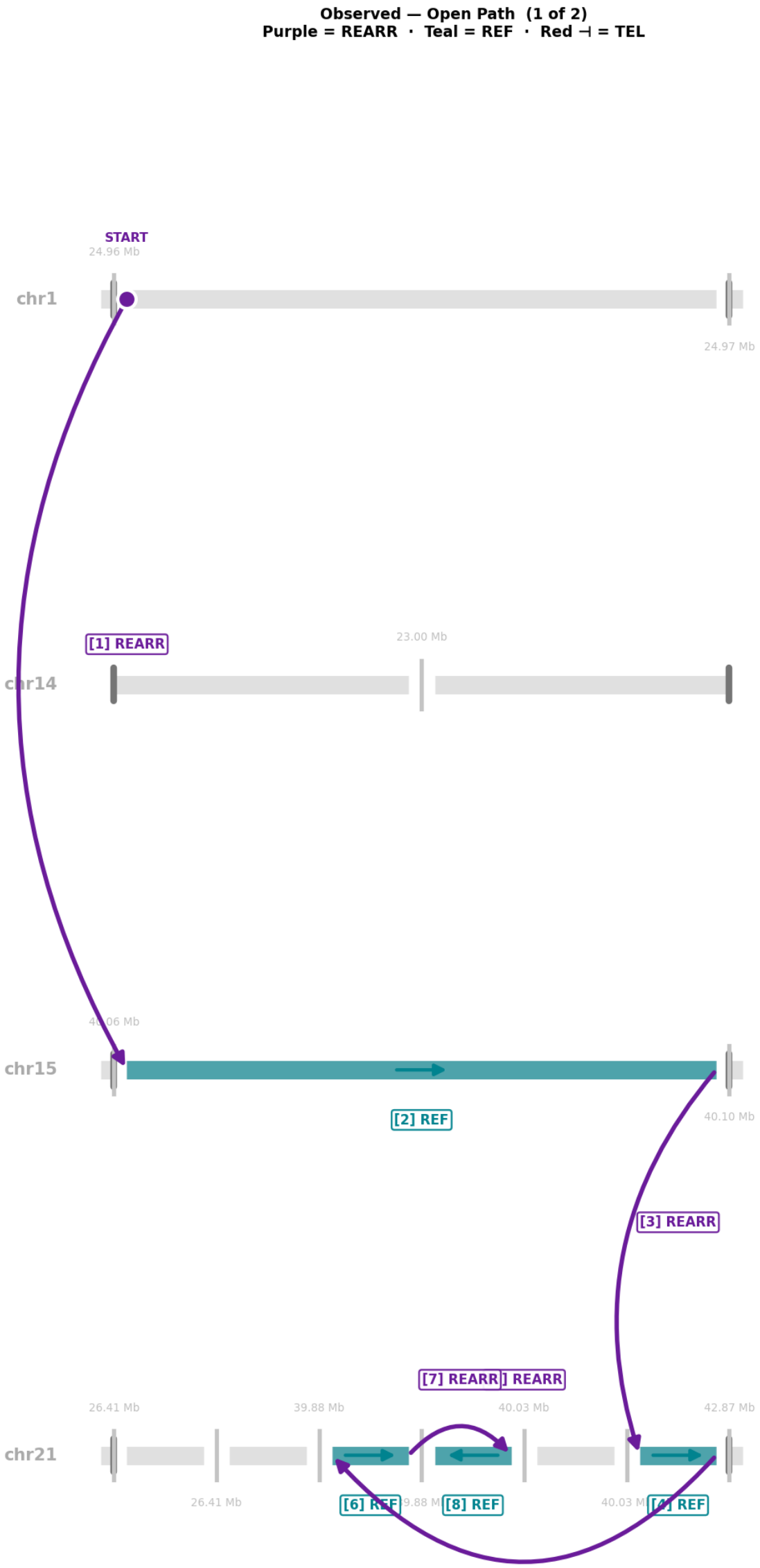
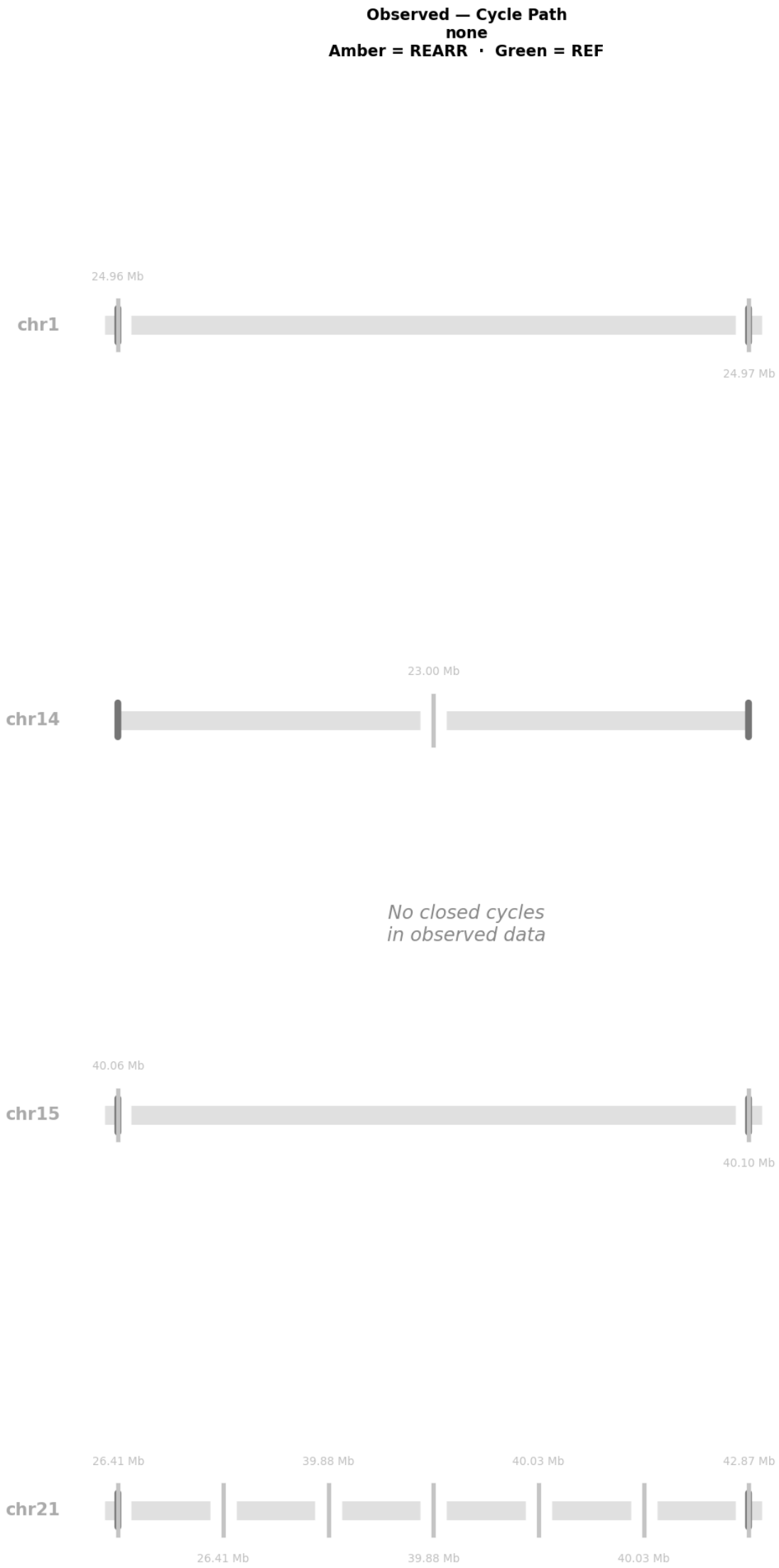
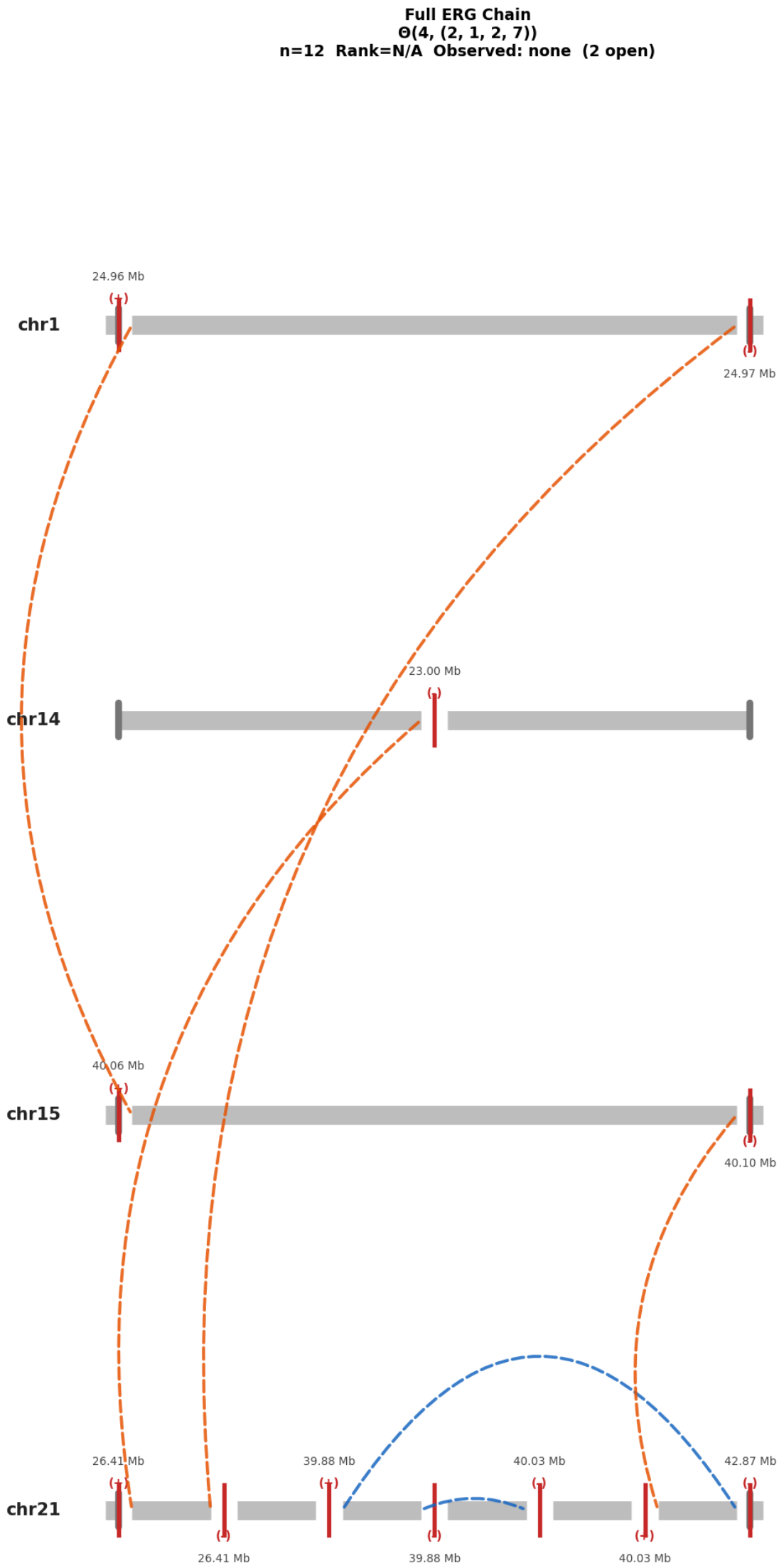


*Correct Repair

*Incorrect Repair

Possible Rejoins

Chromoplexy is Baca's name for this: *one single, coordinated shattering-and-reshuffling event* that scrambles pieces of several chromosomes together at once, rather than a series of separate, one-at-a-time mistakes.



Intrachromosomal
Interchromosomal

REARR hop
REF segment
TELOMERE

#2 What's New

This is our FULL Genome Rearrangement Chain.

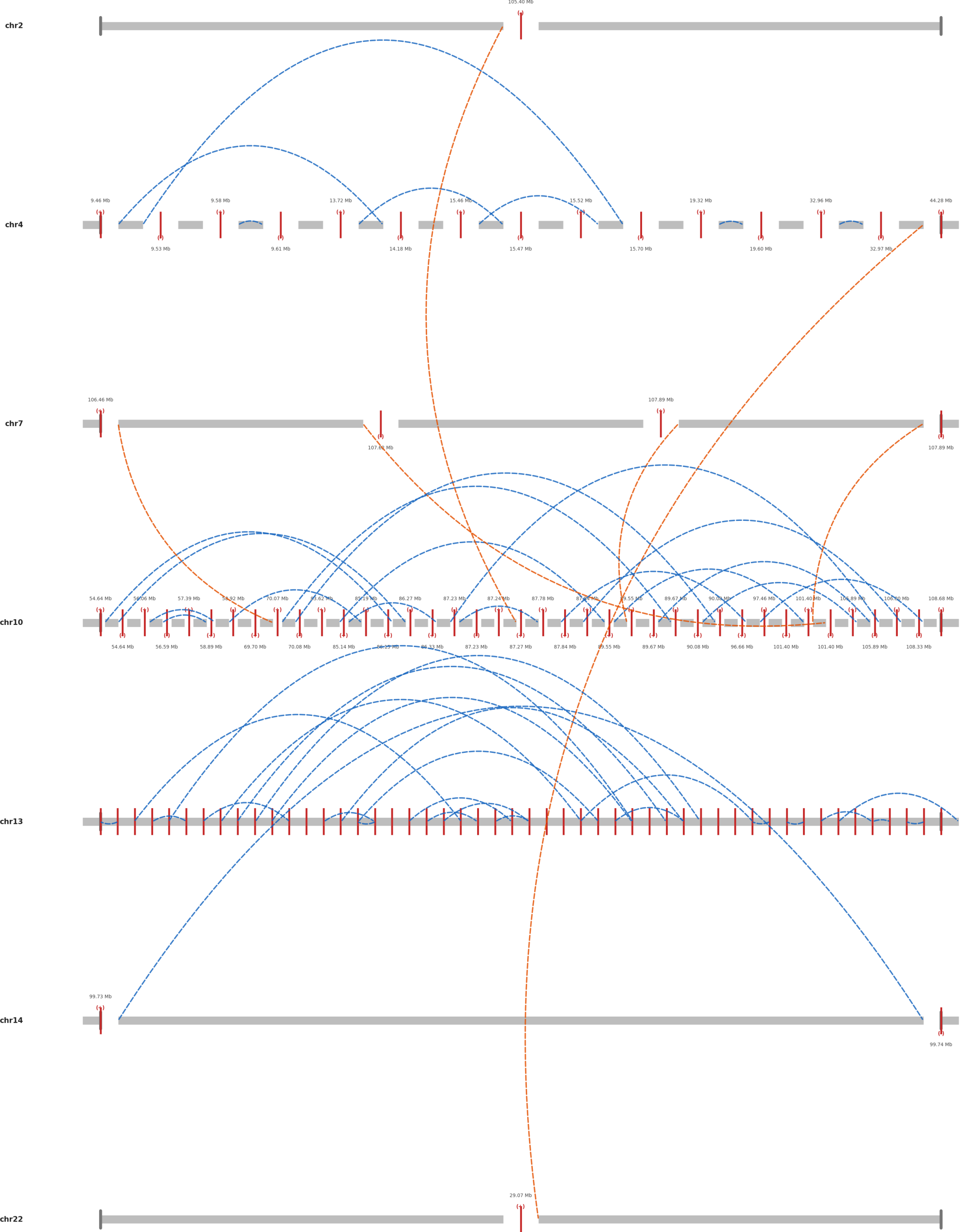
At first, we computed the **ERG chain subset**. This included only the chromosomes **directly** or **one-hop interchromosomally** connected to **chr21** and not chromosomes connected indirectly through another chromosome.

Now, we have the FULL Genome Rearrangement Chain.

Every patient has this kind of chain structure. It incorporates all the chromosomes spanned by the rearrangement data for that patient.

All the **Blue** and **Orange** dashed lines, are accounted for from the **actual rearrangement data** given by Baca.

Now, each patient had one single chain according to us. But, **this is not what Baca proposed**.



Baca's Chains for Each Patient

Baca proposes **multiple** chains for **each** patient.
The **highest number of chains** a patient has is 32!

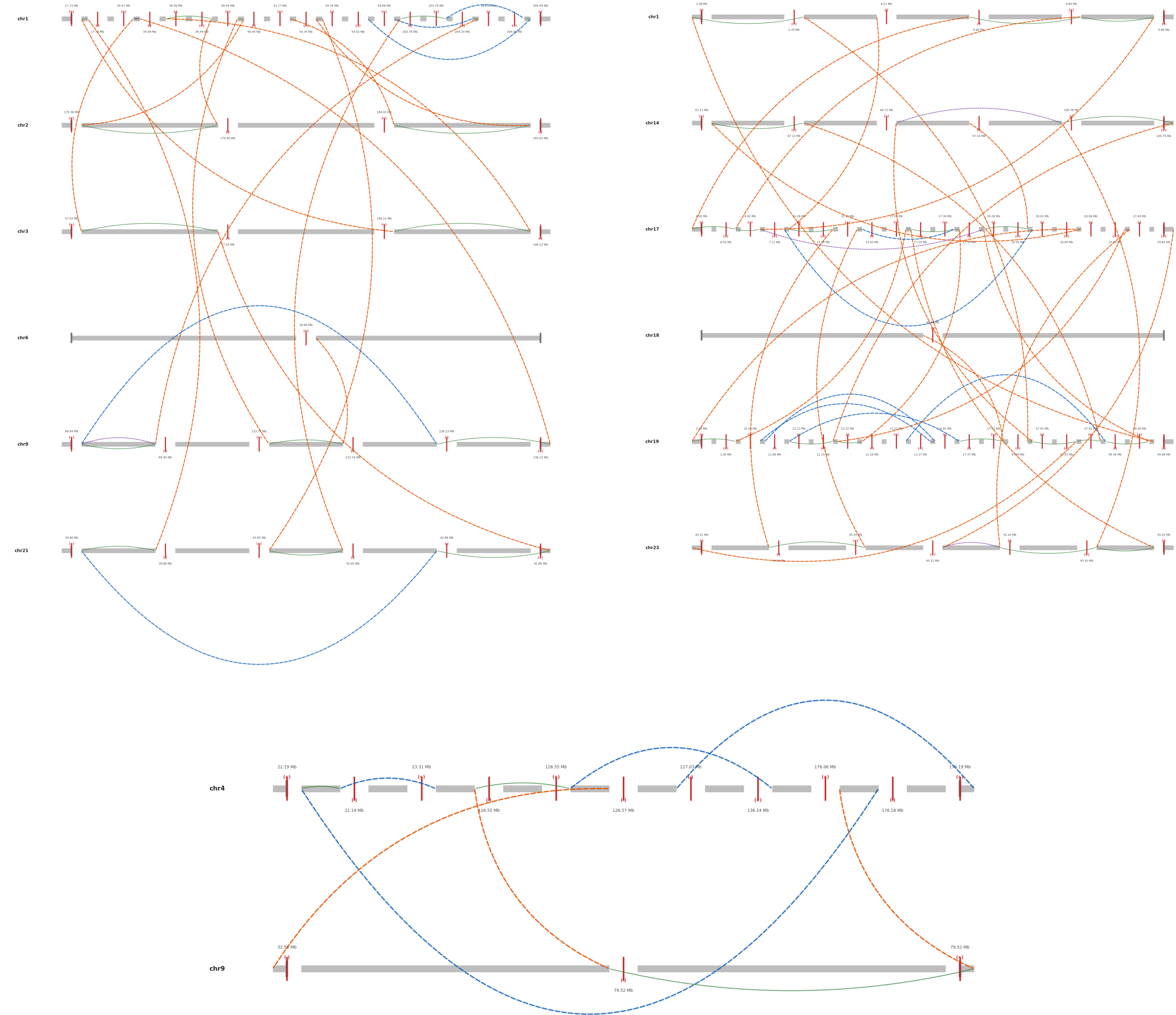
Each chain for each patient spans a different number of chromosomes and distinct chromosomes with minor overlap.

One thing our older chains did not account for was “Deletion Bridges” and “Adjacency Edge” connections.

This gives us Baca's own deletion-bridge and adjacency edges connections we'd otherwise have to guess from genomic position order.

Our own reference-edge rule assumes “next breakpoint by position” is the real neighbor which is **false** whenever a **deletion** sits in between.

ChainFinder statistically rejects the independent-rearrangement hypothesis ($\text{FWER} < 10^{-2}$), we inherit that inference rather than re-deriving it ourselves.



Cycles in Baca’s Chains for Each Patient

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	patient_id	Baca chain number	n breakpoints	n rearrangements	k chromosomes whole chain	chromosomes whole chain	has any	chain full	n open breakpoints	pct breakpoints in closed cycles	n cycles	cycle structure	theta cycles	chromosomes in cycles	max cycle chrom span	has chromoplexy strict	has chromoplexy loose	contains_ERG_or_TMPPRS2	genes_in_chain	ETS_status	Gleason_Score	Pathological_stage
2	P01-28	1	40	20	6	1,2,3,6,9,21	FALSE	FALSE	40	0	0	none	Theta(0,())		0	FALSE	FALSE	TRUE	ABL1,ARHGEF3,CSM	ETS+	4+4	pT3a
3	P01-28	2	20	10	1		FALSE	FALSE	20	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	DNER,EPHA4,PID1,SI	ETS+	4+4	pT3a
4	P01-28	3	14	7	2	4,9	FALSE	FALSE	14	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	ADAM29,GPR125,MIF	ETS+	4+4	pT3a
5	P01-28	4	60	30	6	1,14,17,18,19,23	FALSE	FALSE	60	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	BTBD2,CHD5,CYTSB	ETS+	4+4	pT3a
6	P01-28	5	6	3	2	2,19	FALSE	FALSE	6	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	KCNE4,SGPP2,ZNF4	ETS+	4+4	pT3a
7	P03-1426	1	6	3	1		TRUE	TRUE	0	100	1	C6x1	Theta(1,(6))	2	1	FALSE	FALSE	FALSE	CDKL4,ID2,LOC3397	ETS+	4+3	pT2c
8	P03-1426	2	10	5	2	5,9	TRUE	FALSE	4	60	1	C6x1	Theta(2,(2,4))	5,9	2	FALSE	TRUE	FALSE	FAM169A,FCHO2,GK	ETS+	4+3	pT2c
9	P03-1426	3	6	3	2	19,23	FALSE	FALSE	6	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	EHD2,GLTSCR1,NPA	ETS+	4+3	pT2c
10	P03-1426	4	8	4	2	1,7	FALSE	FALSE	8	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	DHX9,IER5,PPP1R9A	ETS+	4+3	pT2c
11	P03-1426	5	14	7	2	5,9	FALSE	FALSE	14	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	CDK7,CENPH,GPR9	ETS+	4+3	pT2c
12	P03-1426	6	22	11	2	5,18	TRUE	TRUE	0	100	1	C22x1	Theta(2,(6,16))	5,18	2	FALSE	TRUE	FALSE	BASP1,CLUL1,DLGAP	ETS+	4+3	pT2c
13	P03-1426	7	6	3	2	1,13	FALSE	FALSE	0	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	B3GALT,C1QC,IL28F	ETS+	4+3	pT2c
14	P03-1426	8	10	5	2	10,17	TRUE	TRUE	0	100	1	C10x1	Theta(2,(6,4))	10,17	2	FALSE	TRUE	FALSE	LOC100128977,MAR	ETS+	4+3	pT2c
15	P03-1426	9	8	4	3	3,8,15	TRUE	TRUE	0	100	1	C8x1	Theta(3,(2,4,2))	3,8,15	3	TRUE	TRUE	FALSE	CASC5,DLG1,GPR17	ETS+	4+3	pT2c
16	P03-1426	10	6	3	2	1,17	TRUE	TRUE	0	100	1	C6x1	Theta(2,(2,4))	1,17	2	FALSE	TRUE	FALSE	HIC1,ITPKB,METT10C	ETS+	4+3	pT2c
17	P03-1426	11	8	4	1		FALSE	FALSE	8	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	CCDC28A,GPR126,H	ETS+	4+3	pT2c
18	P03-1426	12	6	3	2	7,21	TRUE	TRUE	0	100	1	C6x1	Theta(2,(2,4))	7,21	2	FALSE	TRUE	TRUE	BACE2,ERG,JAZF1,TN	ETS+	4+3	pT2c
19	P03-2345	1	8	4	2	5,21	TRUE	TRUE	0	100	2	C4x2	Theta(2,(2,6))	5,21	2	FALSE	TRUE	TRUE	ERG,SPOCK1,TMPRS	ETS+	3+4	pT2c
20	P03-2345	2	8	4	2	16,24	FALSE	FALSE	8	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	NKD1,PAPD5,UTY,ZN	ETS+	3+4	pT2c
21	P03-2345	3	28	14	2	5,11	TRUE	FALSE	14	50	2	C10x1 + C4x1	Theta(2,(8,6))	5,11	2	FALSE	TRUE	FALSE	AP3B1,ARHGEF12,BU	ETS+	3+4	pT2c
22	P04-1084	1	6	3	1		TRUE	TRUE	0	100	1	C6x1	Theta(1,(6))	11	1	FALSE	FALSE	FALSE	ETS1,LRRC4C	ETS+	3+4	pT3a
23	P04-1084	2	10	5	1		FALSE	FALSE	10	0	0	none	Theta(0,())		0	FALSE	FALSE	TRUE	ADARB1,ERG,TMPRS	ETS+	3+4	pT3a
24	P04-1084	3	8	4	2	17,22	FALSE	FALSE	8	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	CCTB,L2,CRKL,GAB4	ETS+	3+4	pT3a
25	P04-1084	4	10	5	3	2,17,22	FALSE	FALSE	10	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	ARL4C,ATXN10,FLJ4	ETS+	3+4	pT3a
26	P04-1084	5	10	5	3	2,3,8	FALSE	FALSE	10	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	EXO,G,NR4A2,RIPK2	ETS+	3+4	pT3a
27	P04-1084	6	12	6	3	3,8,9	FALSE	FALSE	12	0	0	none	Theta(0,())		0	FALSE	FALSE	FALSE	ARMC1,CNOT10,DIR	ETS+	3+4	pT3a

From the data, about **275** of the chains do **NOT** have **any possible cycle structure**. Meaning, when tracing along the given data, the algorithm could not loop back to any of the breakpoint edges in that chain.

Only **91** chains were found to inhibit cycles.

This does NOT mean that we won’t be able to compute or find cycle structures within the 275 chains. This is where our **cycle completion methods** come in.

The Cornforth Paper

Analyzing Radiation-Induced Complex Chromosome Rearrangements by Combinatorial Painting 1

Author: Cornforth, Michael N.

Source: Radiation Research, 155(5) : 643-659

Published By: Radiation Research Society

URL: [https://doi.org/10.1667/0033-7587\(2001\)155\[0643:ARICCR\]2.0.CO;2](https://doi.org/10.1667/0033-7587(2001)155[0643:ARICCR]2.0.CO;2)

This is the paper used in Sheth et al. (J Phys A) to address the question of **obligate** cycle structures.

The Cornforth Paper

Keyword	Definition
Obligate Cycle Structure	the smallest, most conservative cycle structure consistent with the data: among all valid ways to close the pattern, whichever minimizes the largest cycle. Assumes the least coordination, i.e., breaks rejoined as independently as the data allows.
Probable Cycle Structure	whichever cycle structure occurs most often among every combinatorially possible way the breaks could have rejoined. No conservatism, just raw frequency.

Cornforth's own framing: obligate is what you're forced to conclude at minimum; probable is what's statistically likeliest and the two are often very different structures for the same data.

The Cornforth Paper (Obligate Structure)

Worked examples straight from the paper, confirming the rule:

- Among $\{c_4+c_2, 2c_3\}$ for a 6-break exchange: $2c_3$ has max part 3, c_4+c_2 has max part 4
→ **obligate = $2c_3$** .
- Among $\{c_5+c_3, 2c_4\}$ for an 8-break exchange: $2c_4$ has max part 4, c_5+c_3 has max part 5
→ **obligate = $2c_4$** .
- Among $\{c_6, c_4+c_2, 2c_3\}$ for the paper's own worked Fig. 5 example: $2c_3$ (max 3) beats c_4+c_2 (max 4) and c_6 (max 6) → **obligate = $2c_3$** .

The two-tiered cycle-closure rule

Before you can even ask "**what's the obligate cycle structure,**" you first have to decide **how to close a pattern (cycle)** that has dangling/one-way ends, i.e., *which open ends get connected to which*. Cornforth gives an explicit, two-step rule for this (section 7.2 of the paper).

Rule 1: close the pattern using the **fewest total additional/initial breaks** necessary.

Rule 2 (used only if Rule 1 still leaves more than one option): among those, prefer the closure that involves the **fewest additional chromosomes**.

Completing Cycles using Cornforth's Rules

Obligate Cycle Structure

1. Find every breakpoint still "open" (no real partner recorded) - Cornforth's "one-way" ends.
2. Enumerate every possible way to pair up those open ends.
3. For each pairing, find the resulting cycle structure and the largest chromosome span it creates.
4. Keep only the pairings that minimize that chromosome span (Cornforth's Rule 2).
5. Among those survivors, pick the structure with the smallest largest cycle (Cornforth's core rule - minimize the worst cycle, then the next-worst, and so on).
6. Report that as the obligate (most conservative) structure.

Most Probable Cycle Structure

1. Start from the same open ends.
2. Enumerate every possible pairing - no filtering this time.
3. Compute the resulting cycle structure for each pairing.
4. Count how often each distinct structure shows up.
5. Report whichever structure occurs most frequently as the most probable structure.

Completing Cycles using Cornforth's Rules

n_dangling						e_n_open_path	uses_invent	n_invented	phase2_en	n_matchings	obligate_cycle_structure	obligate_n_c	obligate_ma	obligate_ma	obligate_has	obligate_has	obligate_rule	obligate_n_n	obligate_n_n	probable_cycle_structure	probable_n_c	probable_ma	probable_ma	probable_has	probable_has	probable_pc	probable_n_distinct_s	probable_is
4	2	TRUE	2	TRUE	3	C28x1 + C12x1	2	6	TRUE	TRUE	TRUE	6	3	1	C40x1	1	6	TRUE	TRUE	TRUE	66.67	2	FALSE					
2	1	TRUE	1	TRUE	1	C20x1	1	1	TRUE	FALSE	FALSE	1	1	1	C20x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
8	4	TRUE	4	TRUE	105	C6x1 + C4x1 + C2x2	4	2	TRUE	FALSE	TRUE	2	105	1	C14x1	1	2	TRUE	FALSE	TRUE	45.71	9	FALSE					
14	7	TRUE	7	FALSE																								
2	1	TRUE	1	TRUE	1	C6x1	1	2	TRUE	FALSE	TRUE	2	1	1	C6x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
0	0	FALSE	0	TRUE	1	C6x1	1	1	TRUE	FALSE	FALSE	0	1	1	C6x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C6x1 + C4x1	2	2	TRUE	FALSE	TRUE	2	1	1	C6x1 + C4x1	2	2	TRUE	FALSE	TRUE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C6x1	1	2	TRUE	FALSE	TRUE	2	1	1	C6x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C8x1	1	2	TRUE	FALSE	TRUE	2	1	1	C8x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
6	3	TRUE	3	TRUE	15	C8x1 + C4x1 + C2x1	3	2	TRUE	FALSE	TRUE	2	15	1	C14x1	1	2	TRUE	FALSE	TRUE	53.33	5	FALSE					
0	0	FALSE	0	TRUE	1	C22x1	1	2	TRUE	FALSE	TRUE	0	1	1	C22x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C6x1	1	2	TRUE	FALSE	TRUE	2	1	1	C6x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
0	0	FALSE	0	TRUE	1	C10x1	1	2	TRUE	FALSE	TRUE	0	1	1	C10x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
0	0	FALSE	0	TRUE	1	C8x1	1	3	TRUE	TRUE	TRUE	0	1	1	C8x1	1	3	TRUE	TRUE	TRUE	100	1	FALSE					
0	0	FALSE	0	TRUE	1	C6x1	1	2	TRUE	FALSE	TRUE	0	1	1	C6x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C8x1	1	1	TRUE	FALSE	FALSE	1	1	1	C8x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
0	0	FALSE	0	TRUE	1	C6x1	1	2	TRUE	FALSE	TRUE	0	1	1	C6x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
0	0	FALSE	0	TRUE	1	C4x2	2	2	TRUE	FALSE	TRUE	0	1	1	C4x2	2	2	TRUE	FALSE	TRUE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C8x1	1	2	TRUE	FALSE	TRUE	2	1	1	C8x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C14x1 + C10x1 + C4x1	3	2	TRUE	FALSE	TRUE	2	1	1	C14x1 + C10x1 + C4x1	3	2	TRUE	FALSE	TRUE	100	1	FALSE					
0	0	FALSE	0	TRUE	1	C6x1	1	1	TRUE	FALSE	FALSE	0	1	1	C6x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C10x1	1	1	TRUE	FALSE	FALSE	1	1	1	C10x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
6	3	TRUE	3	TRUE	15	C4x1 + C2x2	3	2	TRUE	FALSE	TRUE	2	15	1	C8x1	1	2	TRUE	FALSE	TRUE	53.33	4	FALSE					
2	1	TRUE	1	TRUE	1	C10x1	1	3	TRUE	TRUE	TRUE	3	1	1	C10x1	1	3	TRUE	TRUE	TRUE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C10x1	1	3	TRUE	TRUE	TRUE	3	1	1	C10x1	1	3	TRUE	TRUE	TRUE	100	1	FALSE					
6	3	TRUE	3	TRUE	15	C6x1 + C4x1 + C2x1	3	3	TRUE	TRUE	TRUE	3	15	1	C12x1	1	3	TRUE	TRUE	TRUE	53.33	5	FALSE					
2	1	TRUE	1	TRUE	1	C8x1	1	2	TRUE	FALSE	TRUE	2	1	1	C8x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
6	3	TRUE	3	TRUE	15	C38x1 + C14x1 + C2x1	3	5	TRUE	TRUE	TRUE	5	5	1	C54x1	1	6	TRUE	TRUE	TRUE	53.33	5	FALSE					
6	3	TRUE	3	TRUE	15	C6x2 + C4x1	3	2	TRUE	FALSE	TRUE	2	15	1	C16x1	1	2	TRUE	FALSE	TRUE	53.33	4	FALSE					
2	1	TRUE	1	TRUE	1	C6x1	1	1	TRUE	FALSE	FALSE	1	1	1	C6x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C28x1	1	2	TRUE	FALSE	TRUE	2	1	1	C28x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
6	3	TRUE	3	TRUE	15	C16x1 + C6x1 + C2x1	3	3	TRUE	TRUE	TRUE	3	15	1	C24x1	1	3	TRUE	TRUE	TRUE	53.33	5	FALSE					
0	0	FALSE	0	TRUE	1	C6x1	1	1	TRUE	FALSE	FALSE	0	1	1	C6x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C8x1	1	1	TRUE	FALSE	FALSE	1	1	1	C8x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C6x1	1	2	TRUE	FALSE	TRUE	2	1	1	C6x1	1	2	TRUE	FALSE	TRUE	100	1	FALSE					
8	4	TRUE	4	TRUE	105	C8x1 + C2x3	4	1	TRUE	FALSE	FALSE	1	105	1	C14x1	1	1	TRUE	FALSE	FALSE	45.71	7	FALSE					
6	3	TRUE	3	TRUE	15	C2x3	3	2	TRUE	FALSE	TRUE	2	5	1	C6x1	1	3	TRUE	TRUE	TRUE	53.33	3	FALSE					
4	2	TRUE	2	TRUE	3	C4x1 + C2x1	2	2	TRUE	FALSE	TRUE	2	3	1	C6x1	1	2	TRUE	FALSE	TRUE	66.67	2	FALSE					
8	4	TRUE	4	TRUE	105	C2x4	4	1	TRUE	FALSE	FALSE	1	105	1	C8x1	1	1	TRUE	FALSE	FALSE	45.71	5	FALSE					
8	4	TRUE	4	TRUE	105	C6x1 + C4x3	4	1	TRUE	FALSE	FALSE	1	105	1	C18x1	1	1	TRUE	FALSE	FALSE	45.71	7	FALSE					
4	2	TRUE	2	TRUE	3	C4x1 + C2x1	2	1	TRUE	FALSE	FALSE	1	3	1	C6x1	1	1	TRUE	FALSE	FALSE	66.67	2	FALSE					
2	1	TRUE	1	TRUE	1	C6x1	1	1	TRUE	FALSE	FALSE	1	1	1	C6x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
2	1	TRUE	1	TRUE	1	C6x1	1	1	TRUE	FALSE	FALSE	1	1	1	C6x1	1	1	TRUE	FALSE	FALSE	100	1	FALSE					
8	4	TRUE	4	TRUE	105	C2x4	4	1	TRUE	FALSE	FALSE	1	105	1	C8x1	1	1	TRUE	FALSE	FALSE	45.71	5	FALSE					

ChainFinder statistically rejects the independent-rearrangement hypothesis (FWER < 10⁻²), we inherit that inference rather than re-deriving it ourselves.

Completing Cycles using Cornforth's Rules

Chains with n_dangling_ends = 0 (already fully closed, no invention needed)	52 / 366
Chains requiring invented closure (n_dangling_ends > 0)	314 / 366
Chains enumerable (n_dangling_ends <= 12)	354 / 366
Chains NOT enumerable (> 12, see 5.4 for the list)	12 / 366
Of 302 enumerable chains requiring invention: obligate == probable for 108, differ for 194 Chains where probable_is_tied	0 / 302
real_has_chromoplexy_strict	4 / 366
obligate_has_chromoplexy_strict	35 / 354 (enumerable only)
probable_has_chromoplexy_strict	39 / 354 (enumerable only)

***has_chromoplexy_strict** flags a chain as *chromoplexy* only if it contains a single closed cycle spanning **≥3 distinct chromosomes**. A cycle touching just 2 chromosomes is treated as a simple reciprocal exchange, not true chromoplexy, strict requires genuine multi-chromosome coordination.